2D CFD Analysis and Aerodynamic Validation of the Flow Past a Square Cylinder at $Re = 13,000$

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Abstract

This technical report presents an Unsteady Reynolds-Averaged Navier-Stokes (URANS) numerical analysis of the flow past a square cylinder at a Reynolds number of $Re = 13,000$ in the ANSYS Fluent environment. Utilizing 4 different turbulence models (Spalart-Allmaras, $k-\omega$, $k-\varepsilon$ and SST $k-\omega$) special emphasis is placed on near-wall mesh resolution (through the y^+ parameter) and temporal discretization to accurately capture the characteristic phenomenon of this case: the vortex shedding. The parameter of interest for this case, the mean aerodynamic drag coefficient, is validated against the experimental benchmark carried out by Norberg in 1993 [1]. In such study, different tests related to the flow and pressure forces were carried out over various rectangular cylinders with variations of their angle of attack ($\alpha = 0 - 90°$) and their sides' legth ratio ($A/B = 1; 1.62; 2.5; 3$) [1].

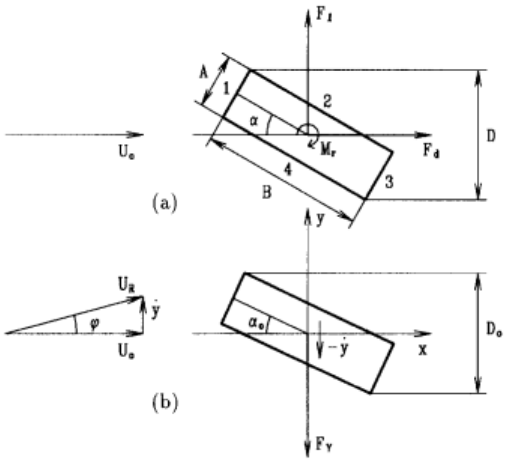


Figure 1: Norberg's experimental workbench [1]

1 Introduction

The flow around a square cylinder is a canonical benchmark in Computational Fluid Dynamics (CFD) due to its fixed separation points at the sharp leading edges and the subsequent complex vortex shedding dynamics. Unlike circular cylinders, where the separation location oscillates due its geometry, the sharp corners of a square cylinder anchor the separation point, making it an ideal test case for validating turbulence models like the already mentioned.

This document outlines the numerical setup, boundary conditions, near-wall treatment and post-processing methodologies implemented to reproduce experimental results for a stationary cylinder of side $L = 0.01$ m at $Re = 13,000$ for $\alpha = 0°$ and $A/B = 1$.

2 Numerical Setup and Methodology

2.1 Governing Equations and Turbulence Models

The simulations were conducted using the Navier-Stokes equations for unsteady and incompressible flow. Such equations are listed below, where Equation 1 represents the Continuity equation and Equation 2 gathers the Momentum equations for the 3 components of the reference frame (Cartesian in this case):

$$\frac{\partial U_j}{\partial x_j} = 0 \tag{1}$$

$$\frac{\partial U_j}{\partial t} + U_j \frac{\partial U_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial p}{\partial x_i} + \nu\left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i}\right) \tag{2}$$

In order to achieve the turbulence closure for the aforementioned equations, the Spalart-Allmaras, Standard $k-\omega$, Standard $k-\varepsilon$ and SST $k-\omega$ model were chosen in order to compare the numerical results and decide which one of them is the most appropiate for this type of analysis.

Commencing with the Spalart-Allmaras model, it was develope by Philippe R. Spalart and Steven R. Allmaras in 1992 [2] and consists of only one equation: the transport equation for $\tilde{\nu}$. This dimensionless variable was originally defined as $\tilde{\nu} = \nu_t / f_{v1}$, where ν_t is the eddy (kinematic) viscosity.

The transport equation for $\tilde{\nu}$ is shown in Equation 3:

$$\frac{\partial \tilde{\nu}}{\partial t} + U_j \frac{\partial \tilde{\nu}}{\partial x_j} = \underbrace{C_{b1} S \tilde{\nu}}_{1} - \underbrace{C_{w1} f_w \left(\frac{\tilde{\nu}}{d}\right)^2}_{2} + \dots \\ \dots + \underbrace{\frac{1}{\sigma}\left[\frac{\partial(\nu+\tilde{\nu})}{\partial x_j}\frac{\partial \tilde{\nu}}{\partial x_j} + C_{b2}\left(\frac{\partial \tilde{\nu}}{\partial x_j}\right)^2\right]}_{3} \tag{3}$$

Where:

1. **Production.** The shear stress generates the turbulence due to velocity gradients.
2. **Destruction.** This term only takes into account $\tilde{\nu}$ destruction caused by non-viscous effects, since the viscous effects are already included in the definition of ν_t. Additionally, when the distance to the wall d approaches zero, $f_{w1} \to 0$ must be satisfied if the $0/0$ singularity is to be avoided.
3. **Diffusion.** The non-linear term $C_{b2}(\partial\tilde{\nu}/\partial x_j)^2$ becomes important in the edges/limits of the turbulent region.

The following turbulence model was the Standard $k-\varepsilon$ model, developed in 1974 by Launder and Spalding [3], in an attempt to determine the Reynolds stresses (and the eddy viscosity), $-\rho\overline{u_i' u_j'}$, after solving the transport equations for the turbulent kinetic energy k and the turbulent dissipation rate ε. Such stresses can be modelled through Boussinesq's hypothesis (4):

$$-\rho\overline{u_i' u_j'} = \mu_t\left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i} - \frac{2}{3}\frac{\partial U_k}{\partial x_k}\delta_{ij}\right) - \frac{2}{3}\rho k \delta_{ij} \tag{4}$$

Where δ_{ij} is the Kronecker delta. The problem that the Boussinesq's hypothesis adds is the necessity of calculating the viscosity μ_t, which can be obtained through different formulations.

The first one is Prandtl's mixing length theory (5):

$$\mu_t = \rho l_m \sqrt{k} = \rho l_m^2 \left|\frac{\partial U}{\partial y}\right| \Leftrightarrow l_m = \kappa y \tag{5}$$

Where $\kappa \sim 0.41$ is the Kármán constant and l_m the mixing length itself. This variable can be defined as the characteristic length that a fluid parcel, such as an eddy, travels before mixing with its surrounding fluid flow and losing its original properties. This explanation can be understood with an easy example: consider two fluid regions, one of them having a temperature $T_1 = T$ and the second of them a temperature $T_2 = 2T$. If an eddy were to move from region T_2 from the region T_1, then l_m would be the required length for the eddy to lower its temperature from $T_2 = 2T$ to $T_1 = T$.

The second formulation that allows the calculation of μ_t is the Van Driest's theory (6):

$$l_m = \kappa y\left[1 - \exp(-y^+/26)\right] \Leftrightarrow y^+ = \frac{\rho y u^*}{\mu} \tag{6}$$

Nevertheless, despite the existence of algebraic expressions for the calculation of μ_t, the most common approach is to solve firstly the transport Equations 7 and 8, and then determine μ_t:

$$\frac{\partial k}{\partial t} + U_j \frac{\partial k}{\partial k} = \frac{\partial}{\partial x_j}\left[\left(\nu + \frac{\nu_t}{\sigma_k}\right)\right] + P_k + P_b - \varepsilon + S_k \tag{7}$$

$$\frac{\partial \varepsilon}{\partial t} + U_j \frac{\partial \varepsilon}{\partial x_j} = \frac{\partial}{\partial x_j}\left[\left(\mu + \frac{\mu_t}{\sigma_\varepsilon}\right)\frac{\partial \varepsilon}{\partial x_j}\right] + \dots \\ \dots + C_1 \frac{\varepsilon}{k}(P_k + C_3 P_b) - C_2 \frac{\varepsilon^2}{k} + S_\varepsilon \tag{8}$$

Thirdly, the $k-\omega$ model was firstly introduced by A. Kolmogorov in 1942 and subsequently refined by D. Wilcox [4] in order to palliate the limitations of the $k-\varepsilon$ model, in particular, its inaccuracy on predicting boundary layers with adverse pressure gradients.

Both dissipation rates, ε and ω (specific rate) are related through Equation 9:

$$\omega = \frac{\varepsilon}{C_\mu k} \approx \frac{9}{100}\frac{\varepsilon}{k} \Leftrightarrow [\omega] = s^{-1}; [\varepsilon] = m^2/s^3 \tag{9}$$

Nonetheless, both magnitudes represent the same phenomenon: the rate or frequency at which the turbulent kinetic energy is transformed into thermal energy.

Like the previous models, the $k-\omega$ model possesses 2 transport Equations (10 and 11):

$$\frac{\partial k}{\partial t} + U_j \frac{\partial k}{\partial x_j} = \tau_{ij}\frac{\partial U_i}{\partial x_j} - \beta^* k\omega + \dots \\ \dots + \frac{\partial}{\partial x_j}\left[(\nu + \sigma^* \nu_t)\frac{\partial k}{\partial x_j}\right] \tag{10}$$

$$\frac{\partial \omega}{\partial t} + U_j \frac{\partial \omega}{\partial x_j} = \alpha\frac{\omega}{k}\tau_{ij}\frac{\partial U_i}{\partial x_j} - \beta\omega^2 + \dots \\ \dots + \frac{\partial}{\partial x_j}\left[(\nu + \sigma\nu_t)\frac{\partial \omega}{\partial x_j}\right] \tag{11}$$

As stated, the $k-\varepsilon$ model shows limitations near walls and requires damping functions that are inaccurate if adverse pressure gradients are present:

$$\begin{aligned} f_1 &= 1 \\ f_2 &= 1 - \frac{3}{10}\exp(-Re^2) \\ f_\mu &= \exp\left[\frac{-3.4}{(1 + Re/50)^2}\right] \end{aligned} \tag{12}$$

The greatest advantage of the $k-\omega$ model is the absence of these functions. On the other hand, this model is very sensitive to the turbulence conditions of the free-stream at the entrance of the dominion: apparently negligible variations of k correspond to considerable changes of μ_t and of the skin friction coefficient c_f. Because of this, the SST $k-\omega$ model was developed in 1993 by Menter [5].

This model combines the formulation of the $k-\omega$ model in the inner parts of the boundary layer (from the wall to the viscous sub-layer) with the behavior of the $k-\varepsilon$ model in the free-stream. Both transport equations, one for the turbulent kinetic energy k and one for the specific turbulent dissipation rate ω, are shown in Equations 13 and 14 respectively.

$$\frac{\partial k}{\partial t}+U_j\frac{\partial k}{\partial x_j}=P_k-\beta^* k\omega+\frac{\partial}{\partial x_j}\left[(\nu+\sigma_k\nu_T)\frac{\partial k}{\partial x_j}\right] \quad (13)$$

$$\frac{\partial \omega}{\partial t}+U_j\frac{\partial \omega}{\partial x_j}=\alpha S^2-\beta\omega^2+\frac{\partial}{\partial x_j}\left[(\nu+\sigma_\omega\nu_T)\frac{\partial \omega}{\partial x_j}\right]+\ldots \cdots+2(1-F_1)\sigma_{\omega 2}\frac{1}{\omega}\frac{\partial k}{\partial x_i}\frac{\partial \omega}{\partial x_i} \quad (14)$$

Where F_1 es known as the bending function and is what allows CFD algorithms to switch between both pre-existing turbulence models: $F_1=0$ for $k-\varepsilon$ and $F_1=1$ for $k-\omega$ (the definition of F_1, alongside the corresponding definitions of the several coefficients present in the already introduced models can be consulted on CFD Online's webpage).

2.2 Computational Domain and Boundary Conditions

The control volume consisted of a velocity inlet located $5L$ upstream, a pressure outlet $15L$ downstream and symmetric side walls positioned $5L$ from the centerline, as shown in Figure 2.

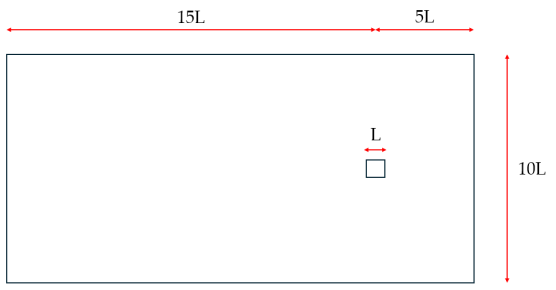


Figure 2: Computational domain

Related to the inlet's velocity condition, aiming to acquire the desired Reynolds number, a velocity magnitude of 19.1 m/s was established, since ISA conditions for air density ($\rho=1.225$ kg/m^3) and dynamic viscosity ($\mu=1.8\cdot10^{-5}$ Pa $\cdot$ s) were considered. Additionally, a turbulence intesity of 5% was introduced.

On the other hand, the gauge pressure of the outlet was set to 0 Pa, which is equivalent to work with an absolute pressure of 101,325 Pa (Note: $p_{abs}=p_{op}+p_{gauge}$).

Lastly, both stationary wall and no-slip conditions were applied to the cylinder itself.

2.3 Mesh Resolution

To properly capture the vortex shedding phenomenon, a family of 4 meshes, from M01 to M08, was developed. Table 1 includes both the number of elements and nodes for each mesh and Figure 3 serves as an illustrative example for the M01 mesh.

Table 1: Mesh family

	M01	**M02**	**M04**	**M08**
$N_{elements}$	31200	61980	124800	250240
N_{nodes}	31670	62644	125740	251571

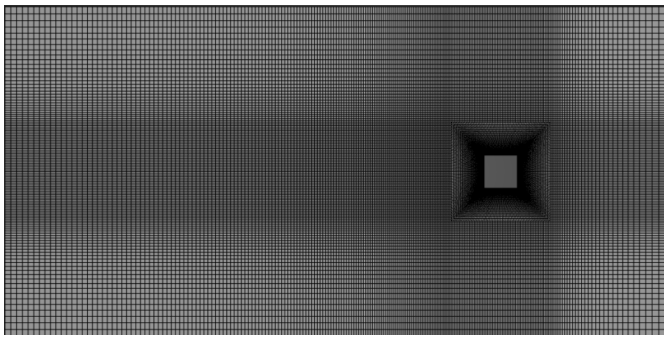


Figure 3: M01

For a subsequent mesh sensitivity analysis, the following refinement criterion was applied to each edge of the computational dominion: let N_i be the number of elements of any edge before the refinement and N_{i+1} after such process, the refinement factor is $r_{i,i+1}=N_{i+1}/N_i=2^{1/D}$, where D is the dominion's dimensionality ($D=2$ for this case). Nevertheless, since this refinement factor is used for 2 different directions, X and Y, the total refinement factor, r_t is of order 2: $r_t\sim r_X\cdot r_Y\sim 2^{1/D}\cdot 2^{1/D}\sim 2^{2/D}\sim 2$.

3 Results and Discussion

Note: before addressing the results of the different simulations, in the Report Definitions section, due to the unusual position of both inlet and outlet for this case, the components of the Force Vector for the drag coefficient must be $(-1,0)$.

The selected solving algorithm was the SIMPLEC one (which stands for Semi-Implicit Method for Pressure Linked Equations Consistent). Nevertheless, to explain the SIMPLEC algorithm it would be best to introduce its little brother: the SIMPLE algorithm.

SIMPLE considers Equations 1 and 2 and makes a prediction on the pressure field, p^*, and uses it to obtain a provisional value for the velocity field, U_j^*. However, this velocity prediction does not satisfy the Continuity equation and corrections on both the pressure and velocity fields, p' and U_j', must be taken. Thus, the correct solution for both fields are $p = p^* + p'$ and $U_j = U_j^* + U_j'$.

The Momentum equations can be rewritten in a semi-discretized way:

$$a_p U_{j,p} = -\sum_n a_n U_{j,n} + \frac{U_j^0}{\Delta t} - \frac{\partial p}{\partial x_j} = H(U_j) - \frac{\partial p}{\partial x_j} \quad (15)$$

Where:

- a_p represents any physical phenomenons that are discretized for the p cell of the computational dominion.
- a_n gathers any physical phenomenons discretized for the neighboring n cells of the p cell.
- $U_j^0/\Delta t$ is the non-stationary and source term.

In the same way, the Continuity equation can be discretized:

$$\frac{\partial U_j}{\partial x_j} = \sum_f \mathbf{n} \cdot U_{j,f} = 0 \quad (16)$$

$$U_{j,f} = \left.\frac{H(U_j)}{a_p}\right|_f - \left(\left.\frac{1}{a_p}\right|_f \left.\frac{\partial p}{\partial x_j}\right|_f\right) \quad (17)$$

Where:

- $\mathbf{n}$ is the normal vector of any face f.
- $U_{j,f}$ is the velocity of face f.

Thus:

$$\frac{1}{a_p}\frac{\partial^2 p}{\partial x_j^2} = \frac{\partial}{\partial x_j}\left[\frac{H(U_j)}{a_p}\right] \quad (18)$$

As the reader might notice, Equation 18 happens to be Poisson's equation for p.

The main problem of the SIMPLE algorithm is that it neglects the term $\sum_n a_n U_{j,n}'$. On the other hand, the SIMPLEC algorithm does not neglect it and establishes that $U_{j,n}' \approx U_{j,p}'$, making the pressure-velocity coupling more robust.

The time-history of the $\overline{c_d}$ was monitored for each of the meshes until a stable periodic state was reached, as it is shown in Figure 4.

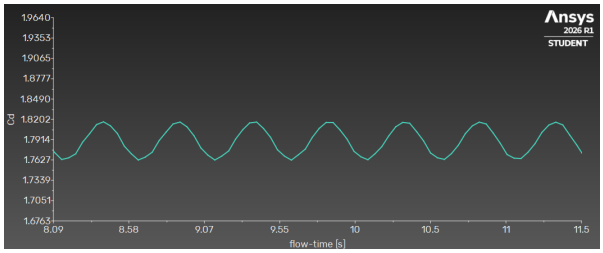


Figure 4: Oscillatory state of $\overline{c_d}$

Once the simulations reached the convergence requirements, the $\overline{c_d}$ values were compared to Richardson's extrapolation, a method for improving numerical results whose main assumption is that the result of a discretised calculation can be expressed in terms of the solutions from the simulations themselves (Equation 19). The main goal of this comparison is estimate an "exact" solution and to quantify the discretization error.

$$f_{\text{Richardson}} = f_1 + \frac{f_1 - f_2}{r^p - 1} \quad (19)$$

Where f_1 and f_2 are the solutions of the fine and coarse meshes, r is the already mentioned refinement factor, p is the order of convergence (calculable using Equation 20) and GCI the Grid Convergence Index related to the fine mesh (21). Also, a Richardson's error, ε_R, can be defined and is shown in Equation 22.

$$p = \frac{1}{\ln r}\ln\left(\frac{f_3 - f_2}{f_2 - f_1}\right) \quad (20)$$

$$\text{GCI } [\%] = \frac{1.25}{r^p - 1}\frac{|f_2 - f_1|}{f_1} \quad (21)$$

$$\varepsilon_R = 100\frac{|f_{\text{Richardson}} - f|}{f_{\text{Richardson}}} \quad (22)$$

Being f_3, f_2, f_1 and f the solution of the coarse, intermediate, fine and treated meshes, respectively. Table 2 shows the selected convergence parameters for the different turbulence models, whereas tables 3, 4, 5 and 6 collect the numerical results of $\overline{c_d}$, ε_R and the maximum value of y^+ on the walls tangent to the flow y^+_{surf}, as well as including a new error, ε_N, which is calculated with the same method as Richardson's error (substituting $f_{\text{Richardson}}$ with Norberg's experimental value of $c_d = 2.16$).

Table 2: Convergence parameters

	S-A	**k − ε**	**k − ω**	**SST k − ω**
ε_R [%]	3.48	5.13	20.94	9.32
GCI [%]	5.38	9.25	56.67	19.98

Table 3: Spalart-Allmaras

	M01	**M02**	**M04**	**M08**
$\overline{c_d}$	1.91	1.98	2.05	2.41
ε_N [%]	11.57	8.33	5.09	11.57
y^+_{surf}	5.2	5.59	3.7	2.93

Table 4: Standard $k-\varepsilon$

	M01	**M02**	**M04**	**M08**
$\overline{c_d}$	1.74	1.79	1.86	2.05
ε_N [%]	19.44	17.13	13.89	5.09
y^+_{surf}	4.84	4.14	2.82	1.93

Table 5: Standard $k-\omega$

	M01	**M02**	**M04**	**M08**
$\overline{c_d}$	1.77	1.79	1.96	2.20
ε_N [%]	18.06	17.13	9.26	1.85
y^+_{surf}	1.32	2.45	3.83	3.7

Table 6: SST $k-\omega$

	M01	**M02**	**M04**	**M08**
$\overline{c_d}$	1.85	1.91	2.01	2.19
ε_N [%]	14.35	11.57	6.94	1.39
y^+_{surf}	2.88	4.82	2.62	3.05

4 Conclusions

- As it can be observed, the Spalart-Allmaras method is the winner in terms of the convergence parameters, while keeping acceptable y^+ values on the walls of the cylinder. However, the model that shows the smallest $\overline{c_d}$ error compared to the experimental result is the SST $k-\omega$ for M08.

- Regardless of the model used, it was possible to capture the vortex street phenomenon for different variables, as it is shown in Figures 5 and 6, which depict the total pressure and velocity magnitude contours of the computational dominion.

- Should the reader be interest in further study of this project, this link will redirect to a GitHub repository that contains this report and the necessary files for the simulations.

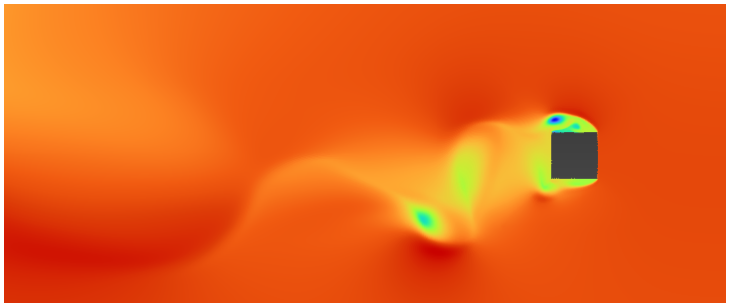


Figure 5: Total pressure shedding

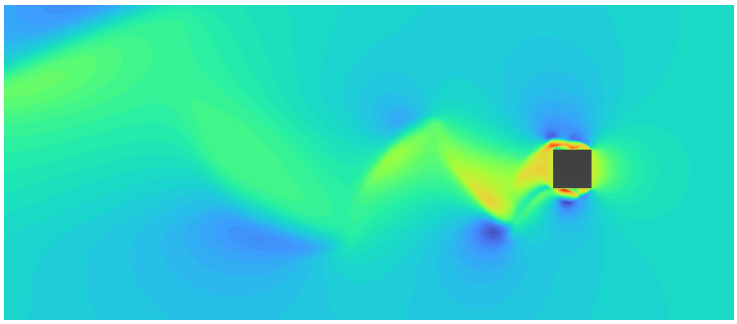


Figure 6: Velocity magnitude shedding

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[1] Norberg, C. (1993). *Flow Around Rectangular Cylinders: Pressure Forces and Wake Frequencies*. Journal of Wind Engineering and Industrial Aerodynamics, Vol. 49, pp. 187-196.

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